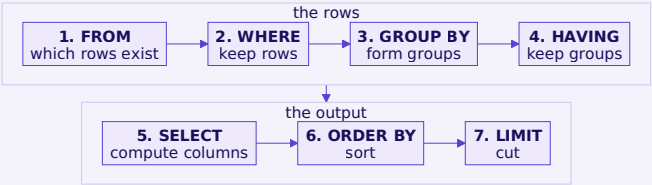


The Monday numbers, in SQL

Kalpa Retail, Week 2 Monday. The revenue tree as queries the warehouse runs every Monday: every count says what it counts, every ratio keeps its fraction, every average names who is in it, and every list has an order.

1 The order a query runs in, and the four lines



Written SELECT first, run FROM first. The run order explains the classic refusals: an alias from SELECT is unknown in WHERE, `WHERE count(*)` is refused, and a column must be grouped or aggregated.

- Count what you mean: `COUNT(DISTINCT customer_id)` counts people; `count(*)` counts rows.
- Divide in numeric and round on purpose: integers divide as integers.
- Name who is averaged: AVG skips NULLs, so say whether a NULL means zero.
- Order every list on a unique key: a table has no order.

2 Week 1's steps as clauses

Week 1 in Python	In SQL
Count the orders	<code>count(*)</code>
Add the amounts	<code>sum(amount)</code>
Count each customer once	<code>count(DISTINCT customer_id)</code>
A total per segment	<code>GROUP BY segment</code>
Keep delivered orders	<code>WHERE status = 'delivered'</code>
The typical order	<code>percentile_cont(0.5) WITHIN GROUP (ORDER BY amount)</code>
Two quarters side by side	Two CTEs lined up on segment

3 Three counts, three questions

Query	Kalpa, two quarters	It answers
<code>count(*)</code> over orders	1,000	How many order rows
<code>count(DISTINCT customer_id)</code> over orders	301	How many customers bought
<code>count(*)</code> over customers	340	How many members Kalpa holds

Crux. A label is a promise the query does not keep: `count(*) AS customers` still counts rows.

4 Division and averages

```
SELECT 140 / 76; -- 1
SELECT round(140::numeric / 76, 2); -- 1.84
SELECT avg(x), count(*), count(x) -- 150, 3, 2
FROM (VALUES (100), (NULL), (200)) AS invented (x);
```

Multiply a ratio back by its denominator: 1.84 times 76 gives 140, and 1 times 76 does not. Put `count(*)` beside every average, and write `coalesce(x, 0)` when no value means zero.

5 WHERE, HAVING and the named step

```
WITH q1 AS (SELECT c.segment, count(*) AS orders
             FROM orders o JOIN customers c USING (customer_id)
             WHERE o.quarter = 'Q1' GROUP BY c.segment),
     q2 AS ( ... the same step, WHERE o.quarter = 'Q2' ... )
SELECT segment, q1.orders, q2.orders
FROM   q1 JOIN q2 USING (segment)
ORDER BY segment;
```

WHERE tests a row before grouping; HAVING tests a group after it. A later step reads every earlier step and every table.

6 The four wrong numbers of the day

The hurried query	The wrong number	The check
<code>count(*)</code> as customers	1,000 customers, 1.00 orders each	Count three ways
<code>LIMIT 5</code> , no ORDER BY	Rs 3,900 audited, Rs 4,590 rerun	Rerun after a reload
<code>count / count</code> in integers	Retail-Plus 2 to 1, "halved"	Multiply back
avg over NULLs	Spend per member 15.5% down	<code>count(*)</code> against <code>count(x)</code>

Crux. Every trap printed a clean number with no error; the check is what catches it.

7 The sentence to Anand

Claim, booked revenue fell 1.6 percent, the rate last week's file showed. Evidence, the rupees fell in Business, 99.1 percent of revenue; the orders and the frequency fell in Retail-Plus, 2.36 to 1.84 orders per customer. Caveat, booked revenue, a thin cell flagged. Next step, collected revenue.

Crux. A number the analyst cannot rerun is a number the analyst cannot sign.

THE WORDS ON THIS SHEET

Warehouse The database every reported number is computed from, queried in place

Schema The tables and columns a database holds, with their types

Query A description of the result you want; the database decides how to fetch it

Logical order The order a query's clauses run in, whatever order they are written in

Distinct count The number of different values in a column

GROUP BY One output row per group, each aggregate computed inside its group

HAVING A filter on groups, after GROUP BY

Integer division Two integers divided to an integer, truncating toward zero